

Candidate Data Transformer — Architecture Design

Deterministic multi-source identity resolution & merge pipeline
Transforms messy multi-source candidate data into a single canonical, explainable, schema-valid candidate profile.

Deterministic Explainable Configurable

IMPLEMENTATION SUMMARY

4
Input Sources

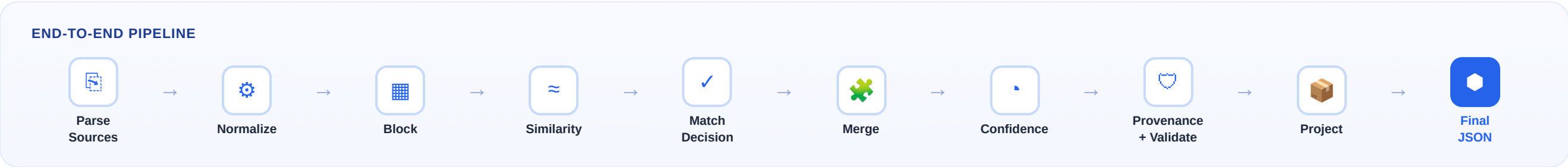
8
Normalizers

9
Pipeline Stages

5
Runtime Configs

6
Test Suites

100%
Deterministic Matching



CANONICAL SCHEMA

Identity id · name	Contact emails · phones
Professional company · title · exp.	Education degree · university · year
Links linkedin · github	Metadata source · raw record
Confidence field + overall score	Provenance sources · reason

NORMALIZATION

Email	Phone	Company	Skills
↓ lowercase	↓ E.164	↓ Alias map	↓ Canonical
Title	Location	Dates	Whitespace
↓ Alias map	↓ Alias map	↓ YYYY-MM	↓ Trim

Fixed, deterministic order — applied uniformly to every source after canonicalization. Invalid emails/phones are dropped, never repaired.

ENGINEERING PRINCIPLES

Deterministic
Same input always produces same output.

Explainable
Every merged field records provenance.

Robust
Malformed data never crashes execution.

Configurable
Output changes without changing pipeline.

Scalable
Blocking minimizes comparisons.

IDENTITY RESOLUTION

- Blocking — email, phone, linkedin, github, last name
- Similarity — exact identifiers + weighted fuzzy match
- Decision — exact ID ⇒ MATCH, else threshold-based
- Output — MATCH / POSSIBLE_MATCH / NO_MATCH

Exact identifiers always dominate; fuzzy similarity only decides ambiguous pairs.

MERGE STRATEGY

- Exact identifier match wins outright
- Else — largest source agreement wins
- Tie → highest source confidence
- Tie → most descriptive value (longest)

Lists (emails, phones, skills) are unioned and deduplicated; nested objects merge field-by-field with the same rule.

CONFIDENCE & PROVENANCE

Source Reliability Cross-source Agreement Field Completeness

↓

Field Confidence → Overall Confidence

Every merged value records its source(s), a confidence score, and the reason it was selected — fully auditable, end to end.

RUNTIME CONFIGURATION

- output_config.json
- Field Selection
- Field Mapping (Rename)
- Confidence / Provenance Toggle
- Missing Value Strategy → Projected JSON

EDGE CASES HANDLED

✓ Duplicate candidates	✓ Invalid email
✓ Invalid phone	✓ Alias drift
✓ Missing values	✓ Conflicting company
✓ Incomplete LinkedIn	✓ Transitive duplicates

OUT OF SCOPE

- ML / probabilistic entity resolution
- Human review workflow
- Live external APIs
- Persistent storage
- Incremental matching

PERFORMANCE & SCALABILITY

- Blocking avoids naive $O(n^2)$ comparison
- Candidates compared only within shared blocks
- Near-linear behavior for realistic dataset sizes